ShakaHub — AI-Generated, Human-Free Educational Social Video Playbook

Honest capabilities, formats, sound, styles, systems, prompts and quality standards

7 October 2026 (research phase)

Contents

How to read this playbook
1. Honest capability research
  1.1 What was actually tested on the box (7 Oct 2026)
  1.2 The proof of concept
  1.3 Capability by category (honest)
  1.4 External tools landscape (verify before relying)
  1.5 Workflow constraints to plan around
2. Format families (what each looks like, and how achievable it is)
  2.1 UI / product videos [NATIVE]
  2.2 Motion graphics [NATIVE]
  2.3 Educational visualisation [NATIVE — Manim/Cairo]
  2.4 Object storytelling
  2.5 Visual metaphors (the core teaching engine)
  2.6 Data storytelling [NATIVE — strongest lane]
  2.7 Cinematic / abstract
3. Sound design (human-free, mostly synthesizable)
  3.1 The four layers
  3.2 Visual action → sound → effect
  3.3 Timeline-based sound design (how the POC was built)
4. Style taxonomy (30 styles)
5. Twenty-five new human-free formats
6. Platform suitability
  6.1 Adapting one idea across platforms ("How compound interest works")
7. Engagement principles without humans
8. Reusable creative system
  8.1 The pipeline: idea → finished video

8.2 Worked example A — “Why do people procrastinate?”
8.3 Worked example B — “Why do startups fail?”
8.4 Worked example C — “How does compound interest work?” (built as the POC)
9. Topic → format decision framework
9.1 Step 1 — classify the knowledge
9.2 Step 2 — topic → format lookup (60 examples)
9.3 Step 3 — feasibility gate
10. Prompt templates
10.1 Master template (copy this)
10.2 Data story (filled: compound interest — the POC)
10.3 Kinetic typography
10.4 UI / product walkthrough
10.5 Visual metaphor / object story
10.6 Educational visualisation (Manim)
10.7 Systems / causal-loop diagram
10.8 Map story
10.9 Hybrid cinematic hook ([EXT] clip + native edit)
11. Quality standards checklist
12. Social Video Capability Map
12.1 The 10 strongest formats for ShakaHub (native, reliable, on-brand)
Appendix — Files

How to read this playbook

This playbook is for making short educational videos for ShakaHub (calm, logical, trustworthy) **with no presenter, no face, no on-camera human**. It is written to be honest: every technique is labelled by *where it can actually be produced*.

Label	Meaning
[NATIVE]	Produced entirely by code on the agent's Linux box (Python + Cairo/Manim/Matplotlib, FFmpeg, procedural audio, local TTS). Deterministic, exact text, frame-accurate sync. **Tested 7 Oct 2026.**
[STILL]	Uses AI still-image generation (GenerateImage) for textures, objects, backgrounds or key art, then animated by code (pan, parallax, masks, reveals).
[EXT]	Requires an external generative tool the user runs/pays for: e.g. Grok Imagine (Grok app / xAI API), Runway, Kling, Veo, Sora for video; Suno/Udio for music; ElevenLabs for premium voice. Availability, tiers, limits and pricing vary and change — verify before planning a series on it.

Bottom line in one paragraph. The strongest, most reliable lane is *code-driven motion graphics with procedurally designed sound*: kinetic typography, charts that grow, counters, UI/product mock-ups, diagrams, maps, flowcharts, isometric/simple 3D, timelines and data stories — all with exact numbers and words, and sound effects locked to the frame. That lane happens to be a perfect fit for a calm, logical, trustworthy brand. Photoreal or "cinematic" footage (macro shots, liquids, fluid 3D worlds, realistic objects moving) is **not** generated natively here; it needs an external text/image-to-video model and should be treated as optional B-roll inserted into a code-built edit.

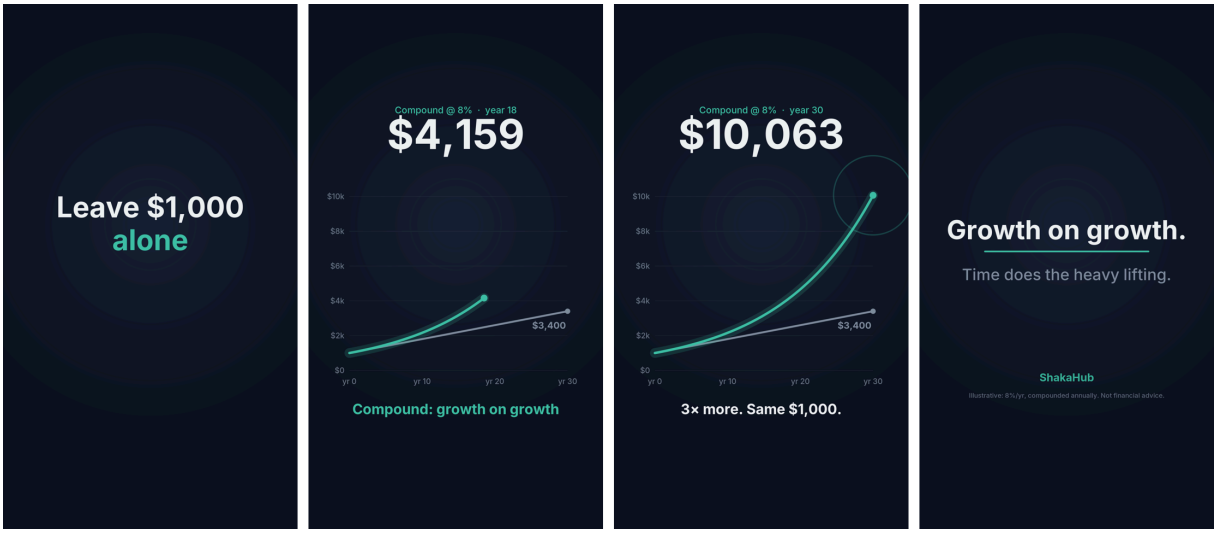
1. Honest capability research

1.1 What was actually tested on the box (7 Oct 2026)

Tool	Status at start	Action	Result
FFmpeg 7.1 (libx264, AAC, loudnorm, ebur128)	Installed	Used for encoding, muxing, loudness	**Works.** H.264 1080×1920 30 fps, AAC 48 kHz, −14 LUFS target
Python 3.13 + Pillow	Installed	—	Works
pycairo (vector 2D renderer)	Missing	Installed system libcairo dev headers + pip in a venv	**Works.** Used for the POC frames (anti-aliased vector text, curves, gradients, glow)
NumPy / SciPy / Matplotlib	Missing	pip in venv	**Works.** SciPy used for filters/convolution reverb in synthesized audio
Manim Community 0.21	Missing	pip in venv	**Works** (rendered an animated exponential-curve test). LaTeX-dependent features (MathTex) need a TeX install — basic one was added, full math typesetting not verified
Node 20 + npm, Google Chrome	Installed	Not exercised	Remotion (React-based video) is *feasible* (Node + headless Chrome present) but **not tested** in this run
Piper neural TTS (en_US "lessac" medium)	Missing	pip + downloaded voice model from Hugging Face	**Works.** Clear, natural-ish neural voice, offline, ~0.5 s to synthesize a sentence
espeak-ng	Missing	apt	Works, but robotic — only for scratch timing
SoX	Missing	apt	Installed; not needed (NumPy synthesis used instead)
Fonts	3,500+ installed incl. Inter, IBM Plex, Manrope, Space Grotesk, JetBrains-style monos	—	Excellent typographic range
Text-to-video diffusion model	**Not available** on the box	—	Cinematic/photoreal motion requires [EXT] tools
Licensed/commercial music library	**Not available**	—	Music must be synthesized, generated externally, or licensed by the user

1.2 The proof of concept

File: `shakahub_compound_interest_poc_9x16.mp4` (1080×1920, 30 fps, 456 frames = 15.2 s, H.264 + AAC, −15 LUFS integrated). A variant without narration: `shakahub_compound_interest_poc_9x16_music_sfx_only.mp4`. Still: `shakahub_poc_still_impact.png`. Source: `poc/` (`timeline.py`, `render.py`, `audio.py`, `sound_cue_sheet.txt`).



POC frames at 1.0 s, 8.3 s, 11.0 s and 14.3 s (hook, chart build, impact, payoff)

Structure (frame-accurate):

Time	Frames	Visual	Sound
0.13–1.2 s	4, 20, 36	Hook words pop in: “Leave $1,000 / alone / for 30 years.”	Click on each word; narration “Leave one thousand dollars alone for thirty years.”
2.13 s	64	“What happens?” fades in	Soft click
2.2–3.0 s	66–90	Hook slides up and out	Rising tone (riser) peaking at frame 88, whoosh peaking at frame 90
3.2 s	96	Grid and axes draw on	Axis click
3.6–6.7 s	108–200	Grey “simple interest” line draws; caption “Simple: a straight line”	Narration: “Simple interest grows in a straight line. Compound interest grows on itself.”
5.0–10.4 s	150–312	Teal compound curve draws with glow; big counter follows the curve head ($1,000 → …), label shows the year	29 year-ticks, each a little higher in pitch (subconscious “growth”)
10.6 s	**318**	Counter lands on **$10,063**, scale pop, ring pulse, micro camera-shake, caption “3× more. Same $1,000.”	**Impact** (sub drop + thump); narration “Ten thousand dollars.”
12.07 s	362	Chart exits; payoff scene	Whoosh
12.2–13.8 s	366–414	“Growth on growth.” → underline draws → “Time does the heavy lifting.” → ShakaHub wordmark + disclaimer	Chime at underline; logo click; narration
14.9–15.2 s	446–456	Fade out	Audio fade

Background bed: soft additive synth pad (Cmaj9 → Am9 → Fmaj7 → Cadd9), low-passed, slowly “breathing”, auto-ducked ~5 dB under narration.

Sync verification: onset detection on the decoded final audio found the click events at 0.13 s and 0.66 s and the impact at 10.60 s — within the 10 ms measurement resolution of their target frames. Sync is exact

by construction: one `timeline.py` file defines every event as a frame number; the renderer and the audio engine both read it, and audio is placed at `sample = frame / 30 × 48000`.

Render cost: ~18 s to render 456 frames of vector graphics on 8 CPU cores; ~3 s for audio synthesis; ~1 s to mux. Iteration is fast — a full series of 30 videos in a templated style is practical.

What it proves: exact numbers, exact on-screen text, data-driven animation, kinetic typography, motion design easing, layered SFX design, synthesized music bed, neural narration, platform loudness — all human-free and reproducible.

What it does *not* prove: photoreal footage, character animation, or "premium" emotional music. Those are [EXT] dependencies.

1.3 Capability by category (honest)

Strong and reliable [NATIVE]

- Kinetic typography, captions, word-by-word reveals, any font, exact spelling.
- Charts and data animations from real numbers (line, bar race, area, pie/donut, waffle, scatter, Sankey-style flows, unit/dot charts).
- Counters, timers, progress bars, gauges, odometers.
- UI/product mock-ups: phones, dashboards, notifications, cursors, typing, toggles, scroll — drawn as vector, crisp at any size.
- Diagrams: flowcharts, systems/causal loops, decision trees, Venn, funnels, timelines, mind-maps, block diagrams.
- Mathematical/educational visualisation (Manim): curves, geometry, probability, exponential growth, distributions.
- Simple 3D: isometric scenes, extruded shapes, rotating cubes/globes, low-poly objects (Manim 3D, or matplotlib 3D, or simple custom projection). Not film-quality lighting.
- Maps: country/region choropleths, animated routes, zooms (needs open GeoJSON data; downloading is possible).
- Procedural sound design: clicks, ticks, whooshes, risers, impacts, chimes, UI blips, typing sounds, heartbeats, drones, glitches, pads; mixing, ducking, reverb, loudness normalisation.
- Neural narration (Piper; quality "good audiobook-lite", one voice tested; others downloadable).
- Batch production from templates (same style, different data/topic), multiple aspect ratios from one timeline.

Possible with care [NATIVE]/[STILL]

- Illustrated scenes using AI-generated stills (objects, paper textures, backgrounds) with parallax, Ken Burns, mask reveals, cut-outs.
- Stop-motion-inspired looks (frame holds at 8–12 fps, jitter, paper textures).
- Pixel art and retro computer looks (fully code-native).
- Hand-drawn/whiteboard "draw-on" effects for vector line art (stroke-dash reveal); hand-drawing photoreal hands is not native.
- Simple object storytelling with flat/2.5D objects (a coin, a battery, a domino chain) built from vector shapes.

Unreliable or not available natively — needs [EXT]

- Photoreal footage: macro shots, liquids, smoke, light rays, realistic materials, camera moves through 3D worlds.
- Consistent characters, lip-sync, facial expression, complex body animation.
- Realistic physics at film quality (falling dominos with real shadows, cloth, water).

- Studio-quality emotive music; celebrity-like or highly expressive voices.
- Watching/critiquing a finished video like a human viewer: the agent cannot watch video; it inspects extracted frames and audio measurements. A human should do the final “feel” check.

1.4 External tools landscape (verify before relying)

- **Grok Imagine (xAI)** — per xAI’s API page and third-party API docs checked 7 Oct 2026: text-to-video and image-to-video, clips of roughly 1–15 s, 480p/720p (1080p on some newer model versions/surfaces), aspect ratios including 9:16, native synchronized audio (SFX/ambience/music/dialogue) that can be switched off, and video edit/extend on the API. Billed per second via API; in the Grok app, access and limits depend on plan. *Treat specifics as changeable.*
- **Runway, Kling, Veo, Sora, Pika, Luma** — similar short-clip generators; strengths differ (camera control, consistency, physics). Clip length is typically 5–15 s; consistency across shots is the main weakness.
- **Suno / Udio** — full music tracks from prompts (licensing terms per plan).
- **ElevenLabs** (and similar) — premium expressive TTS; better than Piper for emotional delivery.

Recommended hybrid: the edit, typography, data, captions, timing and sound design stay [NATIVE] (exact and on-brand); external clips are used only as 2–4 s B-roll or backgrounds, generated with **audio off** so sound design remains under control.

1.5 Workflow constraints to plan around

- The agent can’t watch videos; QA is done via frame grabs, contact sheets, waveform/loudness analysis and onset checks. Keep a human “watch once on a phone” step.
- No licensed music. Options: synthesized beds (native), user-supplied licensed tracks, or [EXT] generated music.
- Render time is not a bottleneck for 2D (≈1–2× real-time or faster); Manim 3D or heavy blur/particle effects can be 10–50× slower.
- Platform uploads/posting are not automated by this playbook; publishing is done by the user or a scheduler.

2. Format families (what each looks like, and how achievable it is)

2.1 UI / product videos [NATIVE]

Vector phone frames, app screens, dashboards, notifications, cursors, taps, scrolls, toggles, typing.

- **Strong uses:** "How it works" for ShakaHub features; habit-tracker streaks; budget screens; "what a good decision dashboard looks like"; before/after settings; onboarding explainers; "3 settings that save you time".
- **Achievable:** pixel-crisp screens, exact copy, animated taps (ripple + click), list reorders, progress rings, charts inside the UI, notification stacks, dark/light themes, device-free floating cards.
- **Unreliable:** photoreal phones in hands, real-world desk shots (→ [EXT] or [STILL] background).
- **Sound:** soft taps, UI blips, keyboard ticks, swipe swishes, success chime, error buzz.

2.2 Motion graphics [NATIVE]

Shapes, lines, icons and type moving with designed easing — the backbone of everything else.

- **Uses:** concept explainers, lists ("3 reasons"), processes, comparisons, definitions.
- **Achievable:** easing (ease-out cubic, back/overshoot, spring), masks, morphs between simple shapes, path draw-on, staggered reveals, parallax layers, grids, loops, particle fields (simple).
- **Unreliable:** organic hand-drawn character acting; complex fluid morphs between arbitrary illustrations.

2.3 Educational visualisation [NATIVE — Manim/Cairo]

Math, science, logic, probability, systems.

- **Uses:** exponential vs linear, base rates, regression to the mean, the Monty Hall problem, the Pareto principle, half-life, sampling error, sunk cost.
- **Achievable:** exact graphs, equations (with LaTeX installed), geometry, number lines, probability trees, dot-arrays (1 dot = 1 person), simulations (random walks, queues) computed in code.
- **Unreliable:** realistic molecular/biological rendering (→ [EXT] or stylised diagrams instead).

2.4 Object storytelling

A single object "lives" a concept: a battery that drains, a coin that splits, a plant that grows, a domino chain, a jar that fills.

- **Achievable [NATIVE]:** flat/2.5D vector objects with squash-and-stretch, fills, cracks, glow; isometric objects; simple physics computed in code (falling, bouncing, stacking).
- **Achievable [STILL]:** AI-generated object stills (e.g. a ceramic piggy bank on a clean background) cut out and animated with transforms, shadows and parallax.
- **Unreliable natively:** photoreal object motion (a real glass shattering, a real plant growing) → [EXT] clip.

2.5 Visual metaphors (the core teaching engine)

Metaphors make abstract ideas concrete. All examples below are buildable [NATIVE] in flat/vector or isometric style; most can be upgraded with [EXT] photoreal clips.

Concept	Visual metaphor	Animation beat
Compound interest	Snowball rolling downhill, growing	Ball radius grows by ×1.08 per "year" tick
Compound interest	Single seed → tree → forest	Branches split recursively each beat
Procrastination	Battery charging "later" while a deadline bar shrinks	Two progress bars racing
Procrastination	Inbox tower stacking higher, wobbling	Blocks drop with thuds, tower sways
Habits	A path worn into grass by repeated footsteps	Line darkens with each pass
Habits	1% better: two lines diverging (1.01^365 vs 0.99^365)	Counter to 37.8 vs 0.03
Startup failure	Leaking bucket (burn rate)	Water level drops, holes appear per cause
Startup failure	Runway: plane accelerating, runway ending	Runway shortens as months tick
Focus	Many browser tabs shrinking a single spotlight	Spotlight radius shrinks per tab
Attention residue	Ghost copies of a previous task trailing behind	Transparent duplicates lag the cursor
Opportunity cost	Fork in the road; second path fades	Unchosen path greys out
Sunk cost	Ship anchored by an ever-growing anchor of "already spent"	Anchor scales up as numbers rise
Inflation	Shopping basket shrinking while price tag stays	Basket scales down per year
Diversification	Eggs moving into many baskets; one basket drops	Only one egg cracks
Risk	Rope bridge with planks missing	Planks fall as risk increases
Bottleneck	Hourglass / funnel with a narrow neck	Particles queue at the neck
Second-order effects	Dominos where each knocks a bigger one	Scale ×1.5 per domino
Network effects	Nodes and links; value = links, not nodes	Link count counter n(n−1)/2
Confirmation bias	Magnifying glass that only highlights matching shapes	Non-matching shapes fade
Survivorship bias	WWII plane with bullet holes; missing planes ghosted	Ghost planes appear
Decision fatigue	Willpower meter draining per choice card	Each card drains 5%
Sleep debt	Bank account in red; nights are deposits	Balance animates
Burnout	Candle burning at both ends	Wax shortens from both sides
Learning curve	Mountain climb with plateau "camps"	Climber dot pauses at plateaus
Dunning–Kruger	Confidence balloon inflating then popping, re-inflating slowly	Balloon scale curve

Concept	Visual metaphor	Animation beat
Feedback loops	Microphone→speaker squeal loop drawn as circular arrows	Loop speeds up
Leverage	Lever and fulcrum; small weight lifts boulder	Fulcrum slides
Signal vs noise	Radio dial tuning out static	Static particles fade
Entropy / clutter	Neat grid of objects slowly randomising	Positions jitter more per beat
Trust	Bridge built plank-by-plank, one bad plank breaks it	Crack sound, plank drops
Time value	Melting ice cube labelled "$100 today"	Ice shrinks
Prioritisation	Jar: big rocks first, then pebbles, then sand	Physics-like drops
Pareto 80/20	20 of 100 dots carry 80% of the weight on a scale	Scale tips
Debt spiral	Whirlpool pulling coins inward	Spiral rotation speeds up
Momentum	Flywheel getting heavier and faster	Rotation speed ramps

2.6 Data storytelling [NATIVE — strongest lane]

- **Uses:** "the number that changes how you see X", bar-chart races, before/after, distributions, "1 dot = 1 person".
- **Achievable:** exact data from CSV/APIs; animated axes; annotated peaks; counters; highlight-and-dim; small multiples; zoom into a data point; source line on screen.
- **Rule for trust:** always show the source and the assumptions (the POC shows "Illustrative: 8%/yr, compounded annually. Not financial advice.").

2.7 Cinematic / abstract

Look	Achievable natively	Unreliable natively (needs [EXT])
Abstract gradients, flowing blobs, light streaks, particle fields, noise fields	Yes — procedural (noise-based shaders in NumPy/Cairo; slower for heavy particles)	—
Depth-of-field look on 2D layers, parallax, slow push-ins	Yes — (blur layers, scale)	—
Geometric "cinematic" scenes: monoliths, grids, horizons, wireframes	Yes — (simple 3D projection/Manim)	Realistic lighting, GI, reflections
Macro shots (ink in water, coins in slow motion, dew on leaves)	No	Yes — Grok Imagine / Veo / Kling / Runway
Fluid 3D worlds, flythroughs, realistic architecture	No (only wireframe/low-poly)	Yes — [EXT]
Realistic people/hands	No (and not needed for this brand)	[EXT], but avoid for "human-free" positioning

Recommended use of cinematic [EXT] clips: 2–3 s mood openers or transitional B-roll, colour-graded to ShakaHub's palette, with all words, numbers and sound added natively on top.

3. Sound design (human-free, mostly synthesizable)

Sound is half of perceived quality. With no presenter, **sound replaces the human energy**: it tells viewers *when* to look and *how* to feel.

3.1 The four layers

1. **Narration (optional)** — Piper neural TTS [NATIVE] or ElevenLabs-class [EXT]. Many top faceless edu videos are text-only + SFX; narration helps retention on TikTok/Shorts but hurts on muted feeds (X, LinkedIn, Facebook) unless captions are burned in.
2. **SFX** — synced to every meaningful visual event [NATIVE, synthesized].
3. **Music bed** — synthesized pads/arpeggios [NATIVE]; licensed or [EXT] generated tracks for richer mood.
4. **Room/air** — very low noise floor or ambience so silence doesn't feel "dead".

Mix targets: speech −16 to −14 LUFS integrated overall; true peak ≤ −1.5 dBTP; music ducked 4–8 dB under voice; SFX transients 3–6 dB above bed.

3.2 Visual action → sound → effect

Visual action	Sound	Effect on viewer	Synth recipe [NATIVE]
Word/card pops in	Short click / tick	Marks rhythm, "something new"	2–3 kHz sine, 6 ms exp decay + noise transient
Number counts up	Rising ticks or rolling "odometer"	Builds anticipation	Ticks with pitch rising per step
Bar/line grows	Rising tone or soft riser	Growth = upward pitch	Sine sweep up 200→700 Hz with swelling gain
Big reveal / final number	Impact (sub drop + thump)	Payoff, memorability	70→38 Hz sine, 350 ms decay + low-passed noise burst
Scene transition	Whoosh	Momentum, novelty	Band-passed noise with sweeping center freq, $\sin^2$ envelope
Before a reveal	Riser, or sudden silence	Tension / pattern interrupt	Ascending tone, then 200–400 ms cut to silence
Correct / success	Two-note chime (major 3rd/5th)	Reward	Additive sines 880 + 1318 Hz, 0.6 s decay
Wrong / myth busted	Low buzz, record-scratch, glitch	Contrast, surprise	Square wave 110 Hz 150 ms; bit-crushed noise
UI tap	Soft tap	Tactile realism	Filtered click, 1.4 kHz, very short
Typing on screen	Keyboard ticks with random spacing	Process, "live" feel	Random-interval micro clicks, varied pitch
Notification appears	Bright blip	Attention	1.8 kHz → 2.4 kHz two-step
Something breaks	Crack / snap	Loss, risk	Noise burst + high-pass, fast decay
Highlight / underline	Soft swish or marker squeak	Focus	Short high band noise swipe
Map zoom / camera push	Low swell + air	Scale, gravity	Low-pass noise swell

Visual action	Sound	Effect on viewer	Synth recipe [NATIVE]
Clock / time passing	Ticking	Urgency	1 Hz click pair
Heartbeat (stress topics)	Double thump	Body/emotion without a human	Two 50 Hz impacts 120 ms apart
Loop end returns to start	Reverse whoosh	Seamless loop	Reversed whoosh buffer

3.3 Timeline-based sound design (how the POC was built)

1. **Write the timeline first** (`timeline.py`): every visual beat is a named frame (`impact=318`).
2. **Visual renderer** reads the same names for animation start times.
3. **Audio engine** converts frames to samples (`frame/FPS*SR`) and places synthesized sounds; transients land exactly on the frame, whooshes are *centered* on the cut, risers *end* at the reveal.
4. **Narration** lines are generated first, their durations measured, and the visual beats are spaced to fit (or the TTS speed `length_scale` is adjusted).
5. **Mix:** SFX get a short convolution reverb for cohesion; music is sidechain-ducked under voice; final loudness normalisation with FFmpeg `loudnorm`.
6. **QA:** cue sheet export (`sound_cue_sheet.txt`) + onset detection on the decoded MP4 to confirm sync.

Rules of thumb

- Not every motion gets a sound. Sound the *meaningful* beats (maybe 1 per 0.5–1.5 s); let small motions be silent.
- Pitch encodes direction: up = growth/good, down = loss/bad.
- Pre-lap: start whooshes ~8 frames before the cut; impacts exactly on the frame; captions 2–4 frames *after* the sound for a "caused by" feel, or on the same frame for punch.
- Silence is a sound — a 300 ms drop-out before the payoff is the cheapest pattern interrupt.
- Keep a consistent "sound palette" per series so ShakaHub becomes recognisable by ear (e.g. the same soft two-note chime as the sonic logo).

4. Style taxonomy (30 styles)

Difficulty is for *this* production setup: ★ easy native, ★★ moderate native, ★★★ hard native / partly [STILL], ★★★★ needs [EXT].

1. Minimalist UI — *Look:* dark/light flat screens, generous spacing, one accent colour. *Best for:* product features, workflows, settings, habits. *Animations:* card slides, taps, toggles, progress rings. *Sound:* taps, blips, soft chime. *Strengths:* very on-brand, crisp, cheap. *Weaknesses:* can feel like an ad. *Difficulty:* ★. *Topics:* "The 2-minute weekly review", "How ShakaHub scores a decision".

2. Infographic — *Look:* icons, numbers, labelled blocks. *Best for:* stats, comparisons, lists. *Animations:* staggered reveals, counters. *Sound:* clicks, ticks, chime. *Strengths:* dense info, skimmable. *Weaknesses:* static if over-designed. ★. *Topics:* "Where your paycheck goes", "Sleep by age".

3. Kinetic typography — *Look:* big type as the main visual. *Best for:* hooks, quotes, definitions, myths. *Animations:* word pops, scale, mask wipes, tracking. *Sound:* click per word, whoosh per line, impact on keyword. *Strengths:* readable muted, fast to make. *Weaknesses:* fatiguing beyond 30 s without visuals. ★. *Topics:* "Stop saying 'I don't have time'", "Words that mean the opposite of what you think".

4. 2D vector illustration — *Look:* flat shapes, simple objects, limited palette. *Best for:* metaphors, processes. *Animations:* transforms, squash/stretch, path draw-on. *Sound:* cartoon-lite foley (pops, boings kept subtle). *Strengths:* friendly, flexible. *Weaknesses:* characters are hard. ★★. *Topics:* "Why habits stick", "How vaccines train immunity".

5. Simple 3D / low-poly — *Look:* extruded shapes, soft shading. *Best for:* objects, spatial ideas. *Animations:* rotations, camera orbit, extrusion growth. *Sound:* deeper whooshes, mechanical clicks. *Strengths:* premium feel. *Weaknesses:* slower render, lighting basic. ★★★. *Topics:* "How a flywheel works", "Why cubes stack and spheres roll".

6. Isometric — *Look:* 30° grid, tiny worlds (city, office, factory). *Best for:* systems, supply chains, organisations. *Animations:* blocks rising, conveyor items, tiles lighting. *Sound:* mechanical ticks, soft bloops. *Strengths:* systems at a glance, satisfying. *Weaknesses:* layout-heavy. ★★. *Topics:* "How a restaurant really makes money", "Where your data goes".

7. Whiteboard — *Look:* black line art on white, hand-drawn feel. *Best for:* step-by-step reasoning. *Animations:* stroke draw-on (dash-offset), arrows. *Sound:* marker squeak, paper. *Strengths:* feels like teaching. *Weaknesses:* dated if slow; no real hand natively. ★★. *Topics:* "Expected value in 30 seconds".

8. Blueprint — *Look:* blue grid, white technical lines, annotations. *Best for:* "how it's built", system design, plans. *Animations:* line draw, dimension markers, zoom to detail. *Sound:* pencil, ruler clicks, low hum. *Strengths:* credibility, calm. *Weaknesses:* niche feel. ★★. *Topics:* "The blueprint of a good morning", "Anatomy of a budget".

9. Technical drawing / exploded view — *Look:* thin lines, callouts, part labels. *Best for:* components, mechanisms. *Animations:* parts separate along axes, callouts appear. *Sound:* mechanical clicks, servo whirs. ★★★. *Topics:* "What's inside a credit score".

10. Data visualisation — *Look:* clean charts, annotations, sources. *Best for:* trends, comparisons. *Animations:* grow, race, highlight, zoom. *Sound:* risers, ticks, impact on key stat. *Strengths:* trust, shareable. *Weaknesses:* needs real data. ★. *Topics:* "Housing cost vs wages since 1980".

11. Editorial — *Look:* serif headlines, rules, pull-quotes, muted palette. *Best for:* analysis, history, explainers. *Animations:* column reveals, highlight sweeps. *Sound:* paper, soft piano-like pad. *Strengths:* authority. *Weaknesses:* less "fun". ★. *Topics:* "Why the 40-hour week exists".

12. Futuristic HUD — *Look:* thin cyan lines, brackets, readouts, scan lines. *Best for:* tech, AI, "inside the system". *Animations:* scanning, targeting brackets, data streams. *Sound:* bleeps, servo, digital hums. *Strengths:* exciting. *Weaknesses:* cliché, can feel less calm. ★★. *Topics:* "How a recommendation algorithm 'sees' you".

13. Paper / card — *Look:* paper textures, cards with soft shadows. *Best for:* lists, stories, timelines. *Animations:* cards flip, slide, stack. *Sound:* paper swipes, card flicks. *Strengths:* tactile, warm. ★ (textures via [STILL] or procedural). *Topics:* "5 cards that explain stoicism".

14. Magazine layout — *Look:* bold grids, big photos (or illustrations), masthead type. *Best for:* stories, profiles of ideas. *Animations:* layout reflows, image wipes. *Sound:* page turns, editorial pad. ★★ (+ [STILL] imagery). *Topics:* "The year minimalism went mainstream".

15. Diagrammatic / systems — *Look:* nodes, arrows, loops. *Best for:* causation, feedback loops. *Animations:* arrows flow, loops pulse. *Sound:* soft blips per node, loop "hum" rising. ★. *Topics:* "The stress–sleep loop".

16. Abstract / generative — *Look:* noise fields, gradients, particles. *Best for:* mood, emotional/philosophical topics, backgrounds. *Animations:* flow, morph, breathe. *Sound:* drones, pads, granular textures. ★★. *Topics:* "What boredom does to your brain" (as backdrop).

17. Cinematic object — *Look:* single realistic object, dramatic light. *Best for:* metaphor hooks. *Animations:* slow push-in, macro. *Sound:* deep impacts, foley. ★★★★ [EXT] (or [STILL] + parallax as a ★★★ substitute). *Topics:* "The hourglass that explains deadlines".

18. Stop-motion-inspired — *Look:* paper cut-outs, held frames (8–12 fps), jitter. *Best for:* playful explainers. *Animations:* stepped motion, small rotations. *Sound:* paper rustle, wooden clicks. ★★. *Topics:* "How a bill becomes a law (cut-out)".

19. Pixel art — *Look:* low-res grid, limited palette. *Best for:* games, systems, nostalgia. *Animations:* sprite steps, tile reveals. *Sound:* chiptune blips (perfectly synthesizable). ★★. *Topics:* "Leveling up habits like an RPG".

20. Retro computer / terminal — *Look:* CRT green/amber, monospace, scanlines. *Best for:* tech history, logic, step-by-step "programs". *Animations:* typing, blinking cursor, ASCII charts. *Sound:* key clicks, beeps, CRT hum. ★. *Topics:* "Your brain's 'if-then' rules".

21. Documentary (faceless) — *Look:* archival-style stills, maps, slow pans, lower thirds. *Best for:* history, events. *Animations:* Ken Burns, map routes, dated captions. *Sound:* ambient drones, low strings-like pads. ★★ (+[STILL]/public-domain images). *Topics:* "The tulip mania in 45 seconds".

22. Scientific — *Look:* plots, lab-notebook annotations, neutral palette. *Best for:* research findings. *Animations:* error bars appear, distributions form. *Sound:* minimal, clean ticks. ★. *Topics:* "What a p-value actually means".

23. Financial terminal — *Look:* tickers, candles, green/red, dense numerics. *Best for:* markets, personal finance. *Animations:* candles build, ticker scroll. *Sound:* ticks, cash-register-lite, impacts. ★. *Topics:* "Why timing the market fails".

24. Architectural — *Look:* plan views, sections, light shadows, line weights. *Best for:* space, structure, "designing your life/week". *Animations:* walls extrude, plan → 3D. ★★★. *Topics:* "Designing a room for

focus".

25. Map-based — *Look:* clean vector maps, routes, choropleths. *Best for:* geography, trade, migration. *Animations:* route drawing, zoom, fill by value. *Sound:* low swells, pins dropping. ★★ (open map data). *Topics:* "Why most of the world's chips come from one island".

New styles proposed for ShakaHub

26. "Calm Lab" — off-white background, graphite lines, single teal accent, lab-notebook annotations. Feels like a quiet scientist thinking. ★. *Topics:* any reasoning topic.

27. Ledger / receipt — thermal-receipt paper, monospace line items printing out. *Animations:* receipt prints line by line, total highlighted. *Sound:* printer whir, tear. ★. *Topics:* "The real cost of a $5 coffee", "What procrastination costs you".

28. Instrument panel / gauges — analog dials, needles, toggles. *Best for:* states (stress, energy, risk). *Animations:* needles swing with overshoot. *Sound:* servo, click detents. ★★. *Topics:* "Your energy gauge across a day".

29. Index cards & strings (evidence board, calm version) — cards connected by thin lines. *Best for:* causes, connected ideas. *Animations:* strings draw between cards. *Sound:* pin taps, string twang. ★★. *Topics:* "Why startups fail: connecting the causes".

30. Topographic / contour — elevation contour lines as metaphor for effort/difficulty. *Animations:* contours ripple, a path climbs. *Sound:* wind, low pad. ★★. *Topics:* "The learning curve is actually a mountain range".

31. Glass/frosted cards (soft UI) — translucent stacked panes, blur, soft light. ★★ (blur is render-heavier). *Topics:* "Three layers of a good decision".

32. Typographic grid poster (Swiss) — strict grid, huge numerals, red/black. ★. *Topics:* "Numbers that matter" series.

5. Twenty-five new human-free formats

All are [NATIVE] unless marked. Durations assume 20–45 s.

1. The Counter That Lies — *Structure:* show a number climbing that viewers assume is good → reveal what it really measures. *Animation:* big odometer, then label flips. *Sound:* rising ticks → record-scratch → calm pad. *Use:* misleading metrics, vanity numbers. *Example:* "10,000 steps… of scrolling." *Holds attention:* expectation set then broken in < 5 s.

2. Two Lines, One Choice — *Structure:* two paths start identical, one small difference, diverge over time. *Animation:* dual line chart with a slider for the difference. *Sound:* ticks per year; impact at final gap. *Use:* habits, saving, learning. *Example:* "1% better vs 1% worse for a year (37.8 vs 0.03)." *Why:* viewers predict the gap and wait to see it.

3. The Receipt — *Structure:* a receipt prints the hidden costs of a choice line by line, ends with total. *Animation:* thermal print scroll. *Sound:* printer whir, tear, cha-ching (soft). *Use:* opportunity cost, subscriptions, time costs. *Example:* "The receipt for 'I'll do it tomorrow'." *Why:* each line is a micro-reveal; total is a payoff.

4. Zoom Out Ladder — *Structure:* start on one tiny fact, zoom out 4–5 levels of scale, each recontextualising. *Animation:* infinite zoom with labelled frames. *Sound:* low swell per level. *Use:* scale, statistics, history. *Example:* "$1 → a day's coffee → a year → a lifetime → invested." *Why:* every zoom is a new curiosity gap.

5. The Dashboard of You — *Structure:* a UI dashboard monitoring a human state (energy, focus, stress) through a day. *Animation:* gauges, alerts, graphs. *Sound:* UI blips, warning tone. *Use:* productivity, wellbeing, ShakaHub product-adjacent. *Example:* "What your focus dashboard looks like after 40 notifications." *Why:* relatable "this is me" + data.

6. Myth → Mechanism — *Structure:* myth in big type, strike-through, then the real mechanism as a diagram. *Animation:* red strike, diagram build. *Sound:* buzz on strike, chime on mechanism. *Use:* misconceptions. *Example:* "Myth: you're lazy. Mechanism: emotion regulation." *Why:* correction is satisfying; clear before/after.

7. Simulated Crowd (1 dot = 1 person) — *Structure:* 100/1,000 dots simulate a population, rules applied, outcome counted. *Animation:* dots move/sort/colour. *Sound:* soft granular patter, impact on result. *Use:* base rates, probability, survivorship. *Example:* "Of 100 startups, here's where they go." *Why:* human scale without humans.

8. The Domino Chain — *Structure:* small cause → chain of effects, each labelled. *Animation:* dominos (vector) falling, growing. *Sound:* each domino click louder/lower. *Use:* second-order effects. *Example:* "How one skipped night of sleep topples your week." *Why:* inevitable momentum — viewers watch to the end of the chain.

9. Build-a-Formula — *Structure:* a formula assembled piece by piece; each variable gets a visual. *Animation:* terms fly in, sliders change values live. *Sound:* click per term, rising tone when value grows. *Use:* finance, science. *Example:* "$A = P(1 + r)^n$, explained by moving one slider." *Why:* interactivity feel; "aha" when slider moves.

10. The Interface Autopsy — *Structure:* a familiar app UI is dissected; each design trick labelled. *Animation:* exploded layers, callouts. *Sound:* scalpel-lite swishes, clicks. *Use:* dark patterns, attention

economy. *Example:* “Why you can’t stop scrolling: 4 design tricks.” *Why:* reveals the hidden; viewers recognise it.

11. Before / After Split — *Structure:* screen split; same scenario two ways; final metric. *Animation:* synchronized halves, wipe. *Sound:* panned left/right SFX. *Use:* decisions, habits, design. *Example:* “Morning with a plan vs without.” *Why:* contrast is hard-wired attention.

12. Timeline Rewind — *Structure:* start at outcome, rewind to the small cause. *Animation:* reverse playback, scrubbing timeline bar. *Sound:* tape-rewind whoosh. *Use:* failures, case studies. *Example:* “This startup died in 2023. Rewind to the decision in 2019.” *Why:* mystery structure.

13. The Scale Test — *Structure:* two things on a balance scale; weights added as arguments. *Animation:* tilting scale with springy physics. *Sound:* weight thuds, creak. *Use:* trade-offs, comparisons. *Example:* “Rent vs buy: putting each factor on the scale.” *Why:* viewers anticipate the tip.

14. Countdown Explainer — *Structure:* “5 → 1” items with a visible countdown ring. *Animation:* ring closes per item. *Sound:* tick-tock, chime at each. *Use:* lists. *Example:* “5 signs your plan is too vague.” *Why:* progress visible = completion drive.

15. Map of a Decision — *Structure:* a decision becomes a tree/map; branches pruned. *Animation:* branches grow, prune with fade. *Sound:* pops, snips. *Use:* reasoning, ShakaHub core. *Example:* “How to decide whether to quit your job — as a map.” *Why:* visible logic = trust.

16. Loading Bar Life — *Structure:* everything framed as loading/progress bars. *Animation:* bars fill at different rates, some stall. *Sound:* progress ticks, stall buzz. *Use:* patience, learning, delayed gratification. *Example:* “Why skills feel stuck at 80%.” *Why:* universal UI language; stalls create tension.

17. The Ghost Path — *Structure:* show what you did (solid) and what you could have done (ghost). *Animation:* transparent alternate path. *Sound:* airy pad for ghost, solid ticks for real. *Use:* opportunity cost. *Example:* “Your ghost portfolio if you’d started at 25.” *Why:* regret/curiosity.

18. Terminal Explains — *Structure:* a retro terminal “runs” a concept as a program. *Animation:* typing, output lines, ASCII bars. *Sound:* key clicks, beep on result. *Use:* logic, algorithms, rules. *Example:* `> run procrastination.exe`. *Why:* novelty + step logic.

19. Object Monologue (text-only) — *Structure:* an object “speaks” via captions (a battery, a credit card). *Animation:* object reacts (blinks, shrinks). *Sound:* object-specific foley. *Use:* personifying concepts without humans. *Example:* “I’m your phone battery. Here’s what you do to me.” *Why:* personality without faces.

20. Fill the Jar — *Structure:* a container fills/empties to show accumulation. *Animation:* particles/marbles drop (simple physics). *Sound:* marble clinks, pouring. *Use:* savings, time budgets, debt. *Example:* “Your 168 hours as a jar of marbles.” *Why:* satisfying ASMR-like motion.

21. The Wrong Graph — *Structure:* show the intuitive (wrong) graph, then morph to the real one. *Animation:* line morph. *Sound:* buzz → whoosh → impact. *Use:* counterintuitive facts. *Example:* “You think happiness rises with income like this… it actually does this.” *Why:* prediction error.

22. Calm Quiz — *Structure:* question, 3 options, 3-second timer, reveal + why. *Animation:* option cards, timer ring. *Sound:* ticking, correct chime. *Use:* engagement, comments. *Example:* “Which grows faster: $100/month at 7% or $200/month at 3%?” *Why:* participation; comment bait without clickbait.

23. Blueprint of a Routine — *Structure:* a routine drawn as an architectural plan with “load-bearing” habits. *Animation:* walls draw, one removed → collapse. *Sound:* pencil, crack. *Use:* habits, systems. *Example:* “The load-bearing habit in your day.” *Why:* elegant metaphor, clear stakes.

24. Signal Meter — *Structure:* claims are tested on a “signal vs noise” meter. *Animation:* needle swings. *Sound:* static to clean tone. *Use:* media literacy, claims checking. *Example:* “Rating 4 viral productivity tips.” *Why:* scoring = curiosity per item.

25. Loop Video (seamless) — *Structure:* ends exactly where it starts so it auto-replays. *Animation:* last frame = first frame; system loop diagram. *Sound:* reverse whoosh into the first beat. *Use:* feedback loops, cycles. *Example:* “The stress loop (watch it repeat).” *Why:* loops boost watch-time/completion on TikTok/Reels.

26. Hybrid Cinematic Hook [EXT + NATIVE] — *Structure:* 2–3 s generated macro clip (e.g. ink drop in water), then native data story. *Animation:* match-cut from ink bloom to chart curve. *Sound:* native SFX, generated audio off. *Use:* premium pieces. *Example:* “This ink drop is how compound interest spreads.” *Why:* high-production first frame for the scroll-stop.

6. Platform suitability

Platform specs and algorithm behaviour change often; treat these as working assumptions for 2026 and check each platform's current guidance before a launch.

Platform	Best aspect / length	Sound behaviour	What works for faceless edu	ShakaHub fit
TikTok	9:16, 15–60 s (longer allowed)	Sound-on culture	Fast hook, narration + SFX, loops, quizzes, myth→mechanism	High reach; keep calm but punchy
Instagram Reels	9:16, 15–45 s	Mixed; many sound-on	Polished motion design, aesthetic consistency, carousel spin-offs	High — brand aesthetic matters
YouTube Shorts	9:16, ≤ 60 s typical (up to 3 min allowed)	Sound-on	Clear explainers, numbers, loops; feeds long-form	High — evergreen search value
X	16:9 or 1:1 (9:16 OK), 15–45 s	Often muted autoplay	Burned-in text, data charts, one strong stat, thread companion	Medium-high for data/finance/tech
Facebook	1:1, 4:5 or 9:16 Reels, 20–60 s	Often muted in feed	Captioned explainers, slower pacing, older audience	Medium
LinkedIn	1:1 or 4:5 (9:16 supported), 30–90 s	Mostly muted	Business/decision content, frameworks, data, calm tone	High — matches "logical, trustworthy"

Safe zones (9:16, 1080×1920): keep key text between ~y 250 and ~y 1550; avoid right ~140 px (TikTok/Reels buttons) and bottom ~350 px (captions/UI). The POC keeps all key elements inside this area.

6.1 Adapting one idea across platforms ("How compound interest works")

Platform	Adaptation
TikTok	15–20 s; narration on; hook "Leave $1,000 alone for 30 years"; seamless loop back to "$1,000"; comment prompt "What would you start with?"
Reels	Same 9:16 master, cleaner end card, cover frame = the $10,063 impact frame; companion carousel of the chart
Shorts	30–40 s version adding "the rule of 72" segment; end card points to a long-form explainer
X	1:1 re-layout (same timeline, different layout constants), no-VO mix, the stat in the first frame, post text with the formula and assumptions
Facebook	4:5, slower pacing (+20% durations), larger type, captions always on
LinkedIn	4:5 or 1:1, framed as "a decision principle: time > timing", add a "What this means for your team/career" scene; no hype

Native advantage: because layout and timing are code, each platform version is a re-render of the same timeline with different layout constants — minutes, not a re-edit.

7. Engagement principles without humans

Principle	What it means	Human-free implementation
Hook in 1–3 s	Give a reason to stay before the thumb moves	First frame already has motion + a specific claim/number (“$1,000 → ?”); no logo intro; sound on frame 1–4
Curiosity gap	Ask a question the viewer wants answered	Show the *question* or the *blank*: “?” counter, covered card, unfinished graph
Progressive reveal	Information arrives in steps	Staggered builds; one idea per beat; reveal the axis before the line
Pattern interrupts	Change every 2–4 s to reset attention	Scene cut, colour flip, zoom, silence drop, style switch (chart → receipt), camera shake on impact
Visual contrast	Differences pop	Grey vs accent lines, before/after split, big vs small type, one colour only for “the answer”
Pacing	Rhythm matches information load	0.5–1.5 s per beat in hooks, 2–4 s for chart reads, hold final payoff ≥ 1.5 s
Text	Readable on a phone, muted	≤ 6–8 words per line, ≥ 48 px body / 90+ px headline at 1080 width, high contrast, burned in
Visual hierarchy	Eye knows where to look	One focal element per moment; dim everything else; motion attracts — move only what matters
Open loops	Promise something later	“The third reason is the one nobody talks about”; partially drawn chart left open
Payoff	Reward for staying	A number, a reframe, or a rule of thumb; plus a short memorable line (“Growth on growth.”)
Loopability	Rewatch	End frame matches start frame; final line leads back into first line
Trust signals	Calm credibility	Sources on screen, stated assumptions, no exaggerated claims, consistent palette and sonic logo

ShakaHub tone guardrails: no fake urgency, no shock imagery, no “they don’t want you to know”. Use curiosity, clarity and satisfying resolution instead of outrage.

8. Reusable creative system

8.1 The pipeline: idea → finished video

1. **Idea** — one question, one answer, one takeaway. Write the takeaway as a ≤ 6-word line.
2. **Format** — choose with the framework in §9 (what kind of knowledge is it?).
3. **Metaphor** — pick one concrete image (§2.5). One metaphor per video.
4. **Scenes** — 3–6 scenes: Hook → Setup → Build (1–3 beats) → Reveal → Payoff/CTA.
5. **Style** — from §4; default ShakaHub house style = "Calm Lab dark": navy #0D1321, off-white #EDEFF3, muted grey #808C9E, accent teal #3FC1A5, Inter.
6. **Pacing** — beat map in frames (30 fps). Hook ≤ 90 frames.
7. **Typography** — Inter Bold for headlines (90–150 px), Inter Regular for support (34–58 px); max 2 sizes per scene.
8. **Transitions** — slide-up + fade (12 frames, ease-out cubic) as default; whip/zoom only for pattern interrupts.
9. **SFX** — map every reveal to §3.2; write the cue sheet.
10. **Music** — synthesized calm pad (default), or [EXT] track; −20 to −24 dB under VO.
11. **Hook** — specific number or counterintuitive claim in frame 1.
12. **Payoff** — a number + a rule of thumb + a short memorable line.
13. **Exact prompt / spec** — fill the template in §10; the agent turns it into `timeline.py` + render + audio scripts.
14. **QA** — §11 checklist; contact sheet + loudness + sync check; one human phone viewing.

8.2 Worked example A — "Why do people procrastinate?"

- **Format:** Myth → Mechanism (+ Two Progress Bars).
- **Metaphor:** a "mood meter" that drops when the task appears; scrolling refills it instantly; the deadline bar keeps shrinking.
- **Scenes:**
    1. Hook (0–2.5 s): kinetic type "It's not laziness." — "laziness" struck through in red-muted.
    2. Setup (2.5–7 s): a task card "Write report" slides in; the mood meter drops 70% (low buzz).
    3. Mechanism (7–15 s): phone UI appears; each scroll tick refills the mood meter (bright blips) while the deadline bar shrinks (clock ticks speed up). Caption: "Your brain is fixing a feeling, not a schedule."
    4. Reveal (15–20 s): diagram: Task → bad feeling → avoid → relief → (loop) — loop arrow pulses.
    5. Payoff (20–26 s): "Shrink the task until the feeling is small." The card splits into "Open doc. Write 1 line." Mood meter barely moves. Chime.
- **Style:** Calm Lab dark + minimalist UI. **Pacing:** 26 s; beats every 1–2 s in hook, 3 s in mechanism.
- **Typography:** Inter Bold 110 px hook; 52 px captions. **Transitions:** slide-up fades; one zoom into the loop.
- **SFX:** strike = buzz; meter drop = descending tone; scroll = blips; clock = ticks accelerating; loop = soft hum; payoff = two-note chime.
- **Music:** low pad in a minor key that resolves to major at the payoff.
- **Hook:** "It's not laziness." **Payoff:** "Make the first step too small to fear."
- **Exact prompt (to the agent):**

> Make a 26 s, 1080×1920, 30 fps human-free explainer "Why do people procrastinate?" for ShakaHub.
> Style: Calm Lab dark (bg #0D1321, text #EDEFF3, grey #808C9E, accent teal #3FC1A5, muted red

> #C46A6A only for the strike), Inter. Scenes: (1) 0–2.5 s kinetic “It’s not laziness.” with “laziness” struck through at 1.6 s; (2) 2.5–7 s task card “Write report” + mood meter drops from 80% to 25%; (3) 7–15 s phone UI scrolling, each of 6 scroll steps refills the meter +10% while a deadline bar shrinks from 7 days to 1; caption “Your brain is fixing a feeling, not a schedule.”; (4) 15–20 s loop diagram Task → bad feeling → avoid → relief → back to Task, arrows pulsing; (5) 20–26 s card splits into “Open the doc. Write 1 line.”; meter dips only 5%; final line “Make the first step too small to fear.” + ShakaHub wordmark. SFX: buzz on strike, descending tone on meter drop, blips per scroll, accelerating clock ticks, hum on loop, chime on final line. Music: soft synth pad A-minor → C-major at 20 s, ducked under VO. Narration: Piper, calm, 4 lines matching scenes. Captions burned in, safe zones respected. Deliver MP4 (−14 LUFS), a cover PNG at the strike frame, and a cue sheet.

8.3 Worked example B — “Why do startups fail?”

- **Format:** Simulated Crowd (1 dot = 1 startup) + Domino Chain.
- **Metaphor:** a leaking bucket of runway; each failure cause is a hole.
- **Scenes:**
  1. Hook (0–3 s): 100 dots appear; “Of 100 startups…” counter; most dots dim one by one quickly (ticks) — “…most don’t make it.”
  2. Setup (3–8 s): bucket labelled “Runway (cash)” with water level; month counter.
  3. Causes (8–22 s): holes open one by one with labels — “No real need”, “Ran out of cash”, “Wrong team”, “Out-competed”, “Pricing/cost” — each hole a drip sound and the water drop speed increases. *(Show cause categories as commonly reported in startup post-mortem surveys; cite the source on screen and avoid precise percentages unless verified.)*
  4. Reveal (22–27 s): the first hole (“No real need”) is shown to cause the others — dominos connect the labels.
  5. Payoff (27–32 s): “Most startups don’t die from one hole. They die from building something nobody needed.” → “Test the need before the product.” Chime + wordmark.
- **Style:** 2D vector + infographic, Calm Lab palette. **Pacing:** 32 s.
- **SFX:** dim ticks, water drips (synth: short resonant sine blips with pitch drop), dripping speeds up, domino clicks, final low impact.
- **Music:** steady low pulse that thins out as water drops; returns full at payoff.
- **Exact prompt:**

> Make a 32 s 1080×1920 human-free explainer “Why do startups fail?” for ShakaHub. Style: 2D vector infographic, Calm Lab dark palette, Inter. Scene 1 (0–3 s): grid of 100 dots, counter “Of 100 startups…”, 90 dots dim in a fast staggered sweep, text “…most don’t make it.” Scene 2 (3–8 s): vector bucket “Runway” with water at 100%, month counter. Scene 3 (8–22 s): 5 holes open sequentially at 2.8 s intervals with labels [No real need, Ran out of cash, Team issues, Out-competed, Pricing/cost], water drains faster after each. Scene 4 (22–27 s): dashed connectors from “No real need” to all other labels (domino logic). Scene 5 (27–32 s): payoff lines and wordmark; source line “Source: <survey name, year>” (verify before publishing). SFX: soft ticks per dot dim, drip per hole, accelerating drips, domino clicks, low impact on payoff. Music: synthesized pulse, thins with water level. Optional Piper narration. Deliver MP4 + cover PNG + cue sheet.

8.4 Worked example C — “How does compound interest work?” (built as the POC)

- **Format:** Two Lines, One Choice + Counter.
- **Metaphor:** growth on growth (the curve feeds itself); optional snowball variant.
- **Scenes:** Hook (kinetic “Leave $1,000 alone for 30 years. What happens?”) → Chart (simple grey line vs compound teal curve, counter following the curve) → Impact ($10,063 vs $3,400) → Payoff (“Growth on growth. Time does the heavy lifting.”).
- **Style:** Calm Lab dark, data visualisation. **Pacing:** 15.2 s (456 frames).

- **SFX:** clicks per hook word, riser + whoosh into chart, axis click, 29 rising-pitch year ticks, impact on final number, whoosh to payoff, chime, logo click.
- **Music:** Cmaj9 → Am9 → Fmaj7 → Cadd9 synth pad, ducked under Piper narration.
- **Exact prompt:** see the full template filled in §10.2 (Data story template); the actual implementation is in `poc/`.

9. Topic → format decision framework

9.1 Step 1 — classify the knowledge

If the topic is mainly…	Ask	Primary format	Secondary format	Style default
A **quantity changing over time**	Is there a real number series?	Data story / Two Lines	Counter That Lies, Ghost Path	Dataviz, Financial terminal
A **comparison**	Two options, one metric?	Before/After Split	Scale Test	Infographic
A **process / sequence**	Ordered steps?	Countdown / step explainer	Terminal Explains, Blueprint	Minimalist UI, Blueprint
A **system / cause-effect**	Loops or chains?	Systems diagram / Domino Chain	Loop Video	Diagrammatic, Isometric
A **misconception**	Is there a wrong intuitive belief?	Myth → Mechanism	The Wrong Graph	Kinetic type + Diagram
A **probability / base rate**	Is it about "how many out of N"?	Simulated Crowd	Calm Quiz	Scientific, Infographic
A **psychological mechanism**	Feeling → behaviour?	Visual metaphor + Dashboard of You	Object Monologue	2D vector, Calm Lab
A **product/workflow**	Is there an interface?	UI walkthrough	Interface Autopsy	Minimalist UI, Glass
A **place / geography**	Does location matter?	Map story	Zoom Out Ladder	Map-based
Scale (very big / very small)	Is magnitude the surprise?	Zoom Out Ladder	Fill the Jar	Isometric, Abstract
History / timeline	Dates and turning points?	Timeline Rewind	Documentary pan	Editorial, Documentary
A **trade-off / decision**	Options with criteria?	Map of a Decision	Scale Test	Diagrammatic
A **formula / rule**	Is there an equation?	Build-a-Formula	Two Lines	Scientific, Blueprint
Mood / philosophy	Is it emotional, not numerical?	Abstract + kinetic type	Hybrid Cinematic Hook [EXT]	Abstract, Editorial
A **definition / single concept**	Can it be said in one line?	Kinetic typography	Object Monologue	Swiss grid

9.2 Step 2 — topic → format lookup (60 examples)

Topic	Format	Metaphor / device	Style
Compound interest	Two Lines + Counter	Growth on growth / snowball	Dataviz
Inflation	Counter That Lies	Shrinking basket	Financial
Emergency fund	Fill the Jar	Jar of months	2D vector
Credit scores	Interface Autopsy	Exploded score card	Technical
Debt avalanche vs snowball	Before/After Split	Two balls rolling	Infographic
Rule of 72	Build-a-Formula	Doubling blocks	Swiss grid
Dollar-cost averaging	Ghost Path	Steps on a staircase	Dataviz
Market timing	The Wrong Graph	Missing best days	Financial terminal
Subscriptions creep	The Receipt	Receipt printout	Ledger
Opportunity cost	Ghost Path	Fork in road	Calm Lab

Topic	Format	Metaphor / device	Style
Procrastination	Myth → Mechanism	Mood meter vs deadline	Minimalist UI
Habit formation	Two Lines	Worn path in grass	2D vector
Decision fatigue	Dashboard of You	Draining meter	UI
Deep work	Loading Bar Life	Spotlight shrinking with tabs	Abstract
Multitasking	Before/After Split	Switching costs as gaps	Diagram
Burnout	Visual metaphor	Candle both ends	2D vector
Sleep debt	Fill the Jar (reverse)	Bank account	Financial
Dopamine & scrolling	Interface Autopsy	Slot machine reels	UI
Confirmation bias	Visual metaphor	Magnifying glass filter	Scientific
Survivorship bias	Simulated Crowd	Plane bullet holes	Documentary
Base rate fallacy	Simulated Crowd	1,000 dots, test results	Scientific
Regression to the mean	The Wrong Graph	Bouncing scores	Dataviz
Sunk cost	Scale Test	Growing anchor	Blueprint
Dunning–Kruger	Two Lines	Balloon	Editorial
Pareto 80/20	Simulated Crowd	Scale with dots	Swiss grid
Second-order thinking	Domino Chain	Growing dominos	Isometric
First principles	Blueprint of a Routine	Breaking into parts	Blueprint
Inversion	Myth → Mechanism	Upside-down map	Calm Lab
Expected value	Build-a-Formula	Coin flips sim	Whiteboard
Probability of streaks	Simulated Crowd	Coin grid	Pixel art
Why startups fail	Simulated Crowd + Domino	Leaking bucket	2D vector
Product-market fit	Signal Meter	Tuning a radio	HUD
Network effects	Systems diagram	Nodes and links	Diagram
Burn rate / runway	Counter + bucket	Runway shortening	Isometric
Pricing psychology	Interface Autopsy	Decoy menu	Editorial
Unit economics	The Receipt	Per-customer receipt	Ledger
Supply chains	Map story	Routes and ships	Map-based
How the internet routes data	Isometric system	Packets as parcels	Isometric
How encryption works	Build-a-Formula	Padlocks and keys	Technical
How AI recommendations work	Interface Autopsy	Funnel of signals	HUD
How LLMs predict text	Terminal Explains	Token probabilities bars	Retro terminal
How GPS works	Map + diagram	Three circles intersecting	Map-based
Why the sky is blue	Educational viz	Light waves scattering	Scientific
Half-life	Educational viz	Dots disappearing	Scientific
Climate & CO_2	Data story	Bathtub inflow/outflow	Dataviz
Population growth	Zoom Out Ladder	Dot field	Map-based
Why cities exist	Isometric system	Roads joining	Isometric
Housing affordability	Two Lines	Wage line vs price line	Dataviz
Interest rates & mortgages	Build-a-Formula	Slider on rate	Financial
Taxes (marginal brackets)	Myth → Mechanism	Buckets filling in order	Infographic
Negotiation anchoring	The Wrong Graph	Anchor drop	Editorial

Topic	Format	Metaphor / device	Style
Learning curve	Topographic	Mountain with camps	Topographic
Spaced repetition	Two Lines	Forgetting curve resets	Scientific
Stoicism control circle	Kinetic type + diagram	Circles of control	Calm Lab
Goal setting	Map of a Decision	Destination + milestones	Map-based
Time management (168 hours)	Fill the Jar	Marbles = hours	Paper/card
Feedback loops (stress)	Loop Video	Circular arrows	Diagram
Trust & reputation	Visual metaphor	Bridge plank by plank	2D vector
Risk vs uncertainty	Calm Quiz	Known vs unknown dice	Infographic
Why diversification works	Simulated Crowd	Eggs & baskets	2D vector

9.3 Step 3 — feasibility gate

- Does the format need photoreal motion? → If yes, either re-express it natively (vector/isometric), or budget an [EXT] clip and keep it ≤ 20% of runtime.
- Is the data verifiable? → If no, use “illustrative” framing and say so on screen.
- Can the takeaway be said in ≤ 6 words? → If no, the topic is two videos.

10. Prompt templates

Use these as briefs to the agent (which converts them into code) or, for [EXT] parts, as prompts to a video generator. Every template has the same 20 fields so briefs are comparable and complete.

10.1 Master template (copy this)

```
TOPIC:          <one question>
AUDIENCE:       <who, what they already know>
OBJECTIVE:      <the one takeaway, <= 6 words> + desired action
HOOK (0-3s):    <exact on-screen words/number + first motion + first sound>
STYLE:          <style # from taxonomy> | palette hex | font family/weights
SCENE-BY-SCENE: S1 <time> <visual> <text> | S2 ... | S3 ... (3-6 scenes)
ANIMATION:      <easing defaults, key motions, what moves vs stays still>
CAMERA:         <static / slow push-in / zoom to detail / shake on impact>
TYPOGRAPHY:     <headline px, body px, max words/line, alignment, caption position>
COLOR:          <bg, text, muted, accent, alert; meaning of each colour>
TRANSITIONS:    <default + exceptions, duration in frames>
SFX:            <event -> sound list, with frames>
MUSIC:          <synth pad / [EXT] track; key, mood, level vs VO, ducking>
NARRATION:      <none | Piper voice | [EXT] voice>; exact lines; speed
PACING:         <beats per second in hook; holds; total beats>
ENDING:         <payoff line, end card, loop or not, CTA>
ASPECT RATIO:   <9:16 1080x1920 | 1:1 1080x1080 | 4:5 1080x1350 | 16:9>
DURATION:       <seconds> @ <fps>
QUALITY:        <H.264 CRF 16-18, -14 LUFS, TP -1.5 dB, safe zones, contrast AA>
AVOID:          <clichés, claims, colours, motions, sounds to avoid>
```

10.2 Data story (filled: compound interest — the POC)

```
TOPIC:          How does compound interest work?
AUDIENCE:       20-35, basic money literacy, sceptical of hype
OBJECTIVE:      "Time does the heavy lifting." -> start early, even small
HOOK:           "Leave $1,000 / alone / for 30 years." word pops at f4/f20/f36 + click each;
                "What happens?" at f64
STYLE:          #10 Data visualisation in Calm Lab dark; bg #0D1321, text #EDEFF3,
                muted #808C9E, accent #3FC1A5; Inter Bold/Regular
SCENE-BY-SCENE: S1 0-3.0s hook | S2 3.0-12.1s chart: grid draws f96, grey simple line
                f108-200, teal compound curve f150-312 with counter tracking the curve
                head, year label | S3 f318 impact $10,063 vs $3,400, caption
                "3x more. Same $1,000." | S4 12.1-15.2s "Growth on growth." underline,
                "Time does the heavy lifting.", wordmark + assumption line
ANIMATION:      ease-out cubic for entrances (8-12 f), back-overshoot for word pops,
                curve draw linear in time (= linear in years), counter = f(curve)
CAMERA:         static; 10-frame decaying micro-shake (7 px) on impact
TYPOGRAPHY:     hook 104 px bold; counter 150 px bold; captions 52 px; labels 28-34 px
COLOR:          teal = compound/answer only; grey = baseline; no red/green
TRANSITIONS:    slide-up 160 px + fade, 10-12 f; whoosh centred on cut
SFX:            click f4,f20,f36; soft click f64; riser ending f88; whoosh f90; axis
                click f96; 29 year ticks rising pitch f155-306; impact f318; whoosh
                f362; chime f390; logo click f414
MUSIC:          synth pad Cmaj9-Am9-Fmaj7-Cadd9, low-passed 1.8 kHz, ~-22 dB,
                ducked 5 dB under VO
NARRATION:      Piper en_US lessac, length_scale 0.92; 4 lines at 0.15/3.35/10.75/12.25 s
PACING:         hook beat every 0.5 s; chart read 7 s; impact hold 1.5 s; payoff 3 s
ENDING:         wordmark + "Illustrative: 8%/yr compounded annually. Not financial advice."
ASPECT RATIO:   9:16 1080x1920 (re-layout constants for 1:1 and 4:5)
DURATION:       15.2 s @ 30 fps (456 frames)
QUALITY:        H.264 CRF 16, yuv420p, AAC 192k, -14 LUFS target, TP -1.5 dBTP
AVOID:          "get rich" language, money-rain clichés, neon green, stock-ticker chaos
```

10.3 Kinetic typography

```
TOPIC:          <a myth, a definition, a quote, a rule>
AUDIENCE:       <scrollers, muted viewing likely>
OBJECTIVE:      One sentence that reframes the topic
HOOK:           First 2-4 words at 120-160 px on frame 1-4, click; key word in accent
STYLE:          #3 Kinetic type or #32 Swiss grid; 1 accent; Inter / Space Grotesk
SCENE-BY-SCENE: S1 hook phrase | S2 the "but" (strike-through or flip) | S3 reason 1-3,
                one line each | S4 reframe line | S5 end card
ANIMATION:      word-by-word pops (back ease), mask wipes for lines, tracking expand for
                emphasis, strike-through draw 8 f
CAMERA:         static; optional 2% slow push-in across whole video
TYPOGRAPHY:     <= 5 words/line, center, line-height 1.1, never smaller than 48 px
COLOR:          bg dark navy, text off-white, accent teal; muted red ONLY for strike
TRANSITIONS:    line push-up 8 f; scene: cut on beat
SFX:            click per word (vary pitch +-5%), whoosh per line change, buzz on strike,
                impact on keyword, chime on reframe
MUSIC:          minimal pulse at 90-100 BPM; place words on beats
NARRATION:      optional; if used, words appear on the frame the word is spoken
                (use TTS phoneme timings or forced alignment)
PACING:         2-3 words per second max on screen
ENDING:         reframe line held 2 s + wordmark; or loop to first word
ASPECT RATIO:   9:16 master; 1:1 for X/LinkedIn
DURATION:       10-25 s @ 30 fps
QUALITY:        no widows, no text in bottom 350 px, contrast >= 7:1
AVOID:          more than 2 fonts, rainbow highlights, shaking text constantly
```

10.4 UI / product walkthrough

```
TOPIC:          <how to do X in ShakaHub / a workflow>
AUDIENCE:       <new users / prospects>
OBJECTIVE:      "Do X in N taps." -> try it
HOOK:           Finished result first (the outcome screen), then "Here's how" rewind
STYLE:          #1 Minimalist UI or #31 Glass; device-free floating phone; brand palette
SCENE-BY-SCENE: S1 outcome screen | S2 tap 1 (ripple) | S3 tap 2 | S4 result animates
                (ring fills / chart updates) | S5 benefit line + end card
ANIMATION:      cursor/tap dot with ripple, list items stagger 3 f, toggles spring,
                number roll-ups, scroll with inertia easing
CAMERA:         zoom to 140% on the tapped element, then back
TYPOGRAPHY:     in-UI text real size; overlay captions 52 px above the phone
COLOR:          UI follows product theme; overlay accent highlights tapped element
TRANSITIONS:    zoom-through the tapped button into the next screen
SFX:            tap per touch, swipe swish, success chime, notification blip
MUSIC:          light, airy pad or soft pluck arpeggio, -24 dB
NARRATION:      optional, one short line per step
PACING:         one action per 1.5-2 s
ENDING:         "N taps. Done." + app name/CTA
ASPECT RATIO:   9:16; 4:5 for LinkedIn
DURATION:       15-30 s @ 30/60 fps (60 fps for scroll smoothness)
QUALITY:        UI copy proofread, real features only, no fake metrics
AVOID:          tiny text, cursor jitter, showing unfinished features as real
```

10.5 Visual metaphor / object story

```
TOPIC:          <abstract concept>
AUDIENCE:       <general>
OBJECTIVE:      The concept felt through one object
HOOK:           The object in an unexpected state (battery at 3%, cracked bridge plank)
STYLE:          #4 2D vector or #6 Isometric; [STILL] upgrade: AI-generated object still,
                cut out, animated with transforms/shadow; [EXT] upgrade: generated clip
SCENE-BY-SCENE: S1 object + question | S2 the force acting on it (labelled) |
                S3 consequence escalates (3 beats) | S4 the fix applied | S5 rule line
ANIMATION:      squash/stretch, fills/drains, cracks drawn as paths, simple physics
                (gravity 9.8 scaled) computed in code
CAMERA:         slow push-in; cut to close-up for the key change
TYPOGRAPHY:     labels attached to object with leader lines; 40-56 px
COLOR:          object in accent; forces in muted grey; danger in muted amber
TRANSITIONS:    match-cut on object shape
SFX:            object foley (synth): drips, cracks, clinks, hums; impact on failure
MUSIC:          tension pad -> resolution chord at the fix
NARRATION:      optional
PACING:         escalate faster each beat (2.5 s, 2 s, 1.5 s)
ENDING:         object restored + rule line
ASPECT RATIO:   9:16
DURATION:       20-35 s
QUALITY:        metaphor must map 1:1 to the concept (no mixed metaphors)
AVOID:          multiple metaphors, gore/violence for "breaking", photoreal hands
```

10.6 Educational visualisation (Manim)

```
TOPIC:          <math/science/logic concept>
AUDIENCE:       <curious non-experts>
OBJECTIVE:      One intuition, e.g. "doubling feels slow, then sudden"
HOOK:           A prediction question with a visible blank ("How many folds to reach
                the Moon? __")
STYLE:          #22 Scientific / #26 Calm Lab; Manim with custom palette
SCENE-BY-SCENE: S1 question | S2 setup objects (axes/shapes) | S3 transform step by step
                | S4 the surprising result | S5 the general rule (formula)
ANIMATION:      Create/Transform/ReplacementTransform; ValueTracker-driven counters;
                highlight with SurroundingRectangle; always_redraw for live labels
CAMERA:         MovingCameraScene zoom to detail; 3D: slow orbit 10-20 deg
TYPOGRAPHY:     MathTex for equations (needs LaTeX), Text (Inter) for words
COLOR:          one colour per variable, consistent throughout
TRANSITIONS:    transforms between states instead of cuts
SFX:            tick per step, rising tone during exponential section, impact on result
MUSIC:          minimal, -26 dB
NARRATION:      recommended (concept-heavy)
PACING:         slow enough to read equations (>= 2 s per equation)
ENDING:         formula + one-line intuition
ASPECT RATIO:   9:16 (config.frame_width/height set for portrait)
DURATION:       30-60 s
QUALITY:        maths checked in code; numbers printed from the same computation
AVOID:          walls of equations, more than 3 colours of variables
```

10.7 Systems / causal-loop diagram

```
TOPIC:          <a loop or chain: stress-sleep, burnout, network effects>
AUDIENCE:       <thinkers, LinkedIn>
OBJECTIVE:      "It's a loop, not a line." + where to break it
HOOK:           A single node pulsing with "Why does this keep happening?"
STYLE:          #15 Diagrammatic / #29 Index cards & strings
SCENE-BY-SCENE: S1 node | S2 arrows extend to 3-5 nodes | S3 loop closes, speeds up |
                S4 a "break point" node highlighted | S5 loop slows, settles
ANIMATION:      arrows draw-on, particles flow along edges, loop speed variable
CAMERA:         start close on node, pull back to reveal whole loop
TYPOGRAPHY:     node labels <= 3 words, 40 px
COLOR:          reinforcing loop = amber, balancing = teal
TRANSITIONS:    none (one continuous diagram)
SFX:            blip per node, hum that rises with loop speed, "snip" at break point
MUSIC:          pulse whose tempo follows loop speed
NARRATION:      optional
PACING:         accelerate then decelerate
ENDING:         settled loop + "Break it here."
ASPECT RATIO:   1:1 or 4:5 for LinkedIn, 9:16 for Shorts
DURATION:       20-40 s
QUALITY:        arrows never cross labels; max 6 nodes
AVOID:          spaghetti diagrams
```

10.8 Map story

```
TOPIC:          <geography-driven fact>
OBJECTIVE:      "Where" explains "why"
HOOK:           Zoom from globe to a single highlighted region with a number
STYLE:          #25 Map-based; muted land, accent for the subject; open GeoJSON data
SCENE-BY-SCENE: S1 globe->region | S2 routes/flows draw | S3 choropleth fill by value |
                S4 zoom to key point | S5 zoom out + takeaway
ANIMATION:      projection interpolation, route dashes moving, pins drop with bounce
CAMERA:         continuous zoom/pan (eased); never more than 3 zoom levels
TYPOGRAPHY:     place labels 32-40 px, data callouts 60 px
COLOR:          sequential single-hue scale; colour-blind safe
TRANSITIONS:    zooms as transitions
SFX:            low swell on zoom, pin drops, route "draw" swish
MUSIC:          documentary-style pad
NARRATION:      recommended
PACING:         2-3 s per zoom
ENDING:         world view + takeaway
ASPECT RATIO:   9:16 or 1:1
DURATION:       20-45 s
QUALITY:        data source on screen; correct borders; projection appropriate
AVOID:          contested borders without care; rainbow colour scales
```

10.9 Hybrid cinematic hook ([EXT] clip + native edit)

Native part: use any template above for the body. **External part** — prompt for Grok Imagine / Veo / Kling / Runway (generate with audio off; 9:16; 4–6 s; then colour-grade natively):

```
Prompt: Macro shot, a single drop of teal ink falling into still clear water against a deep
navy background, the ink blooming outward in slow, branching tendrils that keep doubling,
soft diffused light, shallow depth of field, calm, minimalist, no text, no people, no hands.
Camera: locked-off, very slow push-in. Duration: 5 s. Aspect ratio: 9:16. Audio: off.
Negative: text, logos, people, hands, fast motion, flicker, colour shifts.
```

Then: match-cut the final bloom frame to the chart curve (native), add all text/numbers/sound natively. Expect several generations to get a usable take; check for flicker and artefacts frame-by-frame.

11. Quality standards checklist

Concept & truth

- ☐ One question, one takeaway (≤ 6 words), one metaphor.
- ☐ Every number is computed in code from stated inputs or cited from a named source shown on screen.
- ☐ Assumptions/disclaimers visible where relevant (finance, health).
- ☐ No exaggerated, fear-based or “secret” claims (ShakaHub tone).

Hook & structure

- ☐ Motion + specific claim/number visible by frame 4; first sound by frame 4.
- ☐ Something changes every 2–4 s (pattern interrupts).
- ☐ An open loop is set in the first 3 s and paid off before the end.
- ☐ Payoff held ≥ 1.5 s; end card ≤ 1.5 s.

Visual

- ☐ Resolution and fps exact (e.g. 1080×1920, 30 fps); H.264 yuv420p, CRF 16–18.
- ☐ Key content inside safe zones (y 250–1550, x 60–940 on 9:16).
- ☐ Text ≥ 48 px body, ≥ 90 px headlines; ≤ 8 words/line; contrast ≥ 7:1 for body.
- ☐ One focal point per moment; max 2 type sizes per scene; max 1 accent colour + 1 alert.
- ☐ Consistent easing (no linear “robot” moves except data-driven ones).
- ☐ No jitter, no aliasing on thin lines (≥ 2 px), no banding in gradients (add subtle noise if needed).
- ☐ Spelling/numbers proofread from the source script, not from the render.
- ☐ Contact sheet reviewed (every 1 s) + the cover/thumbnail frame chosen.

Audio

- ☐ Integrated loudness −14 to −16 LUFS; true peak ≤ −1.5 dBTP.
- ☐ Every key visual beat has a matching sound; no sound without a visual cause.
- ☐ Sync verified (cue sheet + onset check; target ≤ 1 frame error).
- ☐ Music ducked under narration; narration intelligible on phone speaker.
- ☐ Works muted: the story is fully understandable from text alone.
- ☐ Only original/synthesized or properly licensed audio.

Platform & brand

- ☐ Version per platform (aspect, length, mix with/without VO).
- ☐ ShakaHub palette, font, sonic logo, wordmark consistent.
- ☐ Captions/burned-in text; alt text and post copy written.
- ☐ Any [EXT] footage disclosed as AI-generated where platform policy requires; checked frame-by-frame for artefacts.
- ☐ Final human check: watch once on a phone, sound on and sound off.

12. Social Video Capability Map

Capability	Possible	Quality	Best use	Limitations	Workflow
Kinetic typography	Yes [NATIVE]	High	Hooks, myths, definitions	Fatigue if >30 s alone	Cairo/Manim → FFmpeg
Animated charts & counters	Yes [NATIVE]	High (exact)	Finance, stats, science	Needs real or labelled data	Data in code → Cairo/Matplotlib/Manim
UI / product mock-ups	Yes [NATIVE]	High	Product, workflows, habits	Not photoreal devices in hands	Vector screens in Cairo; Remotion possible (untested)
Diagrams / flowcharts / loops	Yes [NATIVE]	High	Systems, reasoning	Layout effort for >6 nodes	Cairo/Manim, graph layout in code
Math/science visualisation	Yes [NATIVE]	High	Concepts, intuitions	LaTeX features need full TeX	Manim
Isometric / simple 3D	Yes [NATIVE]	Medium-high	Systems, objects, scale	Basic lighting; slower renders	Custom projection / Manim 3D
Maps	Yes [NATIVE]	Medium-high	Geography, routes	Needs open geodata; projection care	GeoJSON → Cairo
2D vector object stories	Yes [NATIVE]	Medium-high	Metaphors	Characters/acting weak	Cairo shapes + code physics
Pixel art / retro terminal	Yes [NATIVE]	High	Tech, playful series	Niche aesthetic	Cairo/PIL
Illustrated scenes from AI stills	Yes [STILL]	Medium-high	Editorial, documentary, object hero shots	Stills only; motion = pan/parallax/masks; consistency across images	GenerateImage → cut-out → Cairo animate
Procedural SFX	Yes [NATIVE]	Medium-high	All formats	Less organic than recorded foley	NumPy/SciPy synthesis
Synthesized music beds	Yes [NATIVE]	Medium	Calm pads, pulses, chiptune	Not “produced track” quality	NumPy additive/subtractive synth
Narration (neural TTS)	Yes [NATIVE]	Medium	Explainers	Piper less expressive than premium TTS	Piper → measure durations → timeline
Premium voice	[EXT]	High	Flagship videos	Cost, terms, account needed	ElevenLabs etc. → WAV into mix
Licensed/produced music	[EXT] / user	High	Flagship, trending formats	Licensing; not available on box	Suno/Udio or library → mix
Frame-accurate sync & mixing	Yes [NATIVE]	High (≤ 1 frame)	Everything	—	Single timeline file drives video+audio
Multi-platform re-renders	Yes [NATIVE]	High	All platforms	Layout constants per ratio	Same timeline, new layout
Photoreal / macro / cinematic footage	No natively; [EXT]	Varies (often high, inconsistent)	Hooks, B-roll	Cost, generation retries, artefacts, short clips (≈5–15 s)	Grok Imagine / Veo / Kling / Runway / Sora → native edit
Consistent characters & lip-sync	No natively; [EXT] unreliable	Low-medium	Avoid (human-free brand)	Consistency, uncanny	Not recommended

Capability	Possible	Quality	Best use	Limitations	Workflow
Self-review of finished video	Partial	—	QA	Agent can't watch; uses frames + audio analysis	Contact sheet, loudness, onset checks + human view

12.1 The 10 strongest formats for ShakaHub (native, reliable, on-brand)

1. **Data story / Two Lines, One Choice** — growing charts + counters with exact numbers (proven in the POC).
2. **Myth → Mechanism** — kinetic strike-through + causal diagram; perfect for "calm, logical".
3. **Kinetic typography explainer** — fastest to produce, works muted on every platform.
4. **Systems / feedback-loop diagram (incl. seamless Loop Video)** — shows reasoning visibly; strong on LinkedIn and Shorts.
5. **Simulated Crowd (1 dot = 1 person)** — base rates, probability, survivorship with human scale and no humans.
6. **UI / product walkthrough & Dashboard of You** — product-adjacent, pixel-crisp, sets up ShakaHub naturally.
7. **The Receipt / Ledger** — hidden costs, opportunity cost; satisfying line-by-line reveals.
8. **Build-a-Formula (slider explainers)** — rules of thumb made tangible (rule of 72, compound interest).
9. **Visual metaphor object story (2D/isometric)** — leaking bucket, domino chain, fill the jar; the emotional bridge without faces.
10. **Map of a Decision / Scale Test** — decision trees and trade-offs; the clearest expression of ShakaHub's purpose.

Optional premium layer: the **Hybrid Cinematic Hook** ([EXT] 3–5 s generated clip + native body) for flagship posts once a video-generation tool and budget are confirmed.

Appendix — Files

- `shakahub_compound_interest_poc_9x16.mp4` — POC with narration + SFX + pad.
- `shakahub_compound_interest_poc_9x16_music_sfx_only.mp4` — same, no narration (for muted-first platforms).
- `shakahub_poc_still_impact.png` — still frame at the impact (frame 330).
- `poc/timeline.py`, `poc/render.py`, `poc/audio.py`, `poc/sound_cue_sheet.txt` — reproducible source.
- Reproduce: `cd poc && ../.venv/bin/python render.py && ../.venv/bin/python audio.py`, then mux with FFmpeg (see `build.sh`).